

```
SELECT incident_id, root_cause, resolution
FROM engine_failure_history
ORDER BY embedding <-> '[new_incident_embedding]'
LIMIT 3;
```

3. *Preventive action triggered:* One match involved a fuel control unit fault that escalated to engine shutdown mid-air. Based on match confidence and metadata, the airline flags the aircraft for preventive maintenance before the next flight.

Real advantages in aviation safety

Feature	Benefit
Semantic similarity of fault patterns	Captures meaning across wording variations
Low-latency Postgres querying	Fast enough for maintenance triage
Hybrid filtering	Combines structured filters (aircraft_type = 'A320')
Cost-effective and on-premises	Meets airline data compliance needs

Example output (simplified)

Incident ID	Match %	Root cause	Action taken
ENG-2047	92%	FCU calibration drift	Engine swapped
ENG-1879	89%	Vibration sensor under-reading	Sensor replaced
ENG-1935	84%	Compressor stall in early stage	Cleaned and recalibrated

Safety and compliance alignment

- Data stays within airline infrastructure (critical for DGCA/EASA compliance).
- Enables building an airline-specific semantic failure library.
- Reduces false positives by combining sensor + log embeddings.

Use case 2: AI-assisted diagnosis support in rural health centres

Problem: In many rural or Tier-2/Tier-3 areas of India, Primary Health Centres (PHCs) often operate without specialist doctors. Health workers or junior doctors have to diagnose and treat patients based on limited experience and documentation. And the consequences are misdiagnosis or late diagnosis, inconsistent treatment, and increased mortality in emergency or complex cases.

Solution: Semantic case retrieval using pgvector in PostgreSQL.

How it works:

1. Historical medical case data (symptoms, treatments, outcomes) from urban hospitals and research institutions are stored in a PostgreSQL database.
2. Each case is converted into a vector embedding using a language model trained on Indian medical context (e.g., BioBERT, IndicBERT).
3. When a new patient arrives at a rural PHC, their symptoms are entered (in simple English or regional language, which is translated and embedded).
4. A semantic similarity query is run using pgvector to retrieve similar past cases, diagnoses, and outcomes.

```
SELECT case_id, likely_diagnosis, treatment
FROM historical_cases
ORDER BY symptoms_vector <-> '[new_patient_vector]'
LIMIT 3;
```

Why this use case is possible only with pgvector in PostgreSQL/hybrid database

Challenge	Traditional SQL	Vector DBs only	PostgreSQL + pgvector
Complex filtering by region, age, etc	Yes	No	Yes
Real-time, offline query in rural areas	No	No (Cloud-based)	Yes (Runs locally)
Semantic match between cases	No	Yes	Yes
One system for structured + vector data	No	No	Yes
Cost-effective for government roll-out	No (Needs multiple tools)	No (SaaS expensive)	Yes (Open source, local)

Real impact in India:

- Enable frontline health workers to consult an AI-backed system that suggests possible diagnoses and treatments.
- Use regional and public health data to make localised decisions (e.g., 'dengue vs chikungunya' symptoms in Kerala).
- Reduce referral load on tertiary hospitals.
- Train junior medical staff by exposing them to similar cases.